



PROJECT CHARTER

Team: Gameplay — Software Analysis and Development Technologist (SENA)

1. Project Name

Suggested name (pending confirmation by the full team):

"CodeScape: The Developer's Enigma"

Note: the name has not yet been officially defined by the project team. This is an initial suggestion, aligned with the programming and scape room theme; it may be adjusted before final delivery.

2. Sponsor, Project Manager, and Client

Role	Name / Entity
Sponsor	SENA
Project Manager	Juan Esteban Montoya
Client	Instructor Yuely Adriana Arce

3. Responsible Parties and Their Roles

Project team structure, with the person responsible, members, and main function:

Role / Team	Responsible	Members	Main Function
Scrum Master	Juan Esteban Montoya	-	Coordinates the team, manages the backlog, facilitates Scrum ceremonies, and acts as Project Manager.
Game Design	Leandro Calvo	Felipe, Montoya, Samuel	Designs the narrative, levels, rules, and progression of the game.
UX/UI	Joseph Gómez	Vinasco, Sánchez, Stiven, Samuel León	Designs the interface, user experience, and accessibility of the product.
Unity (Team 1)	Luiyer Gamalie	Karen, Marín, Lemus	Development of the game in the Unity engine.
Unity (Team 2)	Michael Ocampo	Jualian, Upegui, Sebastián Sánchez	Development of the game in the Unity engine.
Gameplay	Michael Isaza	Cristians Marmolejo, Santiago Galindo, Dylan Cardona	Designs and implements the game mechanics, difficulty, and level progression.
Backend	Thomas	Dani, Villota, Almario	Development of the server, database, and server-side logic.

Note: within the Gameplay team (Michael Isaza, Cristians Marmolejo, Santiago Galindo, and Dylan Cardona), internal roles among its members have not yet been defined.

4. Start Date and End Date

Start Date	End Date
06/08/2026	19/11/2026

5. Problem Statement

Currently, many people who are starting out in software development find programming concepts abstract and unengaging to learn through traditional methods, such as theoretical classes or lengthy documentation. Within the scope of this course, there is no playful, interactive, and accessible tool that allows users to practice software development concepts and stages (documentation, design, development, and testing) in an entertaining way, while also being inclusive for people with disabilities (no hands, deaf, or blind) and measurable through difficulty levels and scores.

The project aims to address this need by designing and developing a digital scape room, available in web and mobile versions (Android and iOS), with a theme related to programming and software development, aimed at users over the age of 14.

6. Description of Objectives

General objective:

To design and develop a digital scape room (web and mobile) about software development, with a minimum of 6 levels, aimed at users over the age of 14, that allows users to practice programming concepts in a playful way, with different difficulty levels and a scoring system.

Specific objectives:

- Design the project's brand identity (name, logo, and slogan), aligned with the requested color palette (pink, black, blue).
- Design and develop a minimum of 6 levels with a linear storyline and increasing difficulty (easy, intermediate, expert).
- Implement 100% interactive game mechanics, with a time limit to escape, autosave, and a scoring system (historical and per game session).
- Ensure the game can be used by people with no hands, who are deaf, or who are blind.
- Develop the product in a web version and in mobile applications for Android and iOS.
- Enable matches of 1 to 4 players per team.
- Prepare the required documentation: in-game instructions, user manual, installation manual, maintenance manual, mockups, and prototypes.
- Reach a minimum of 1000 users within the scope of the project.

7. Scope

IN SCOPE	OUT OF SCOPE
• Design and development of a digital scape room with a minimum of 6 levels and progressive difficulty (easy, intermediate, expert). • Web version and mobile applications (Android and iOS). • Scoring system: highest historical score and highest score per game session. • Autosave system; progress resets upon losing. • Accessibility for people with no hands, who are deaf, or who are blind.	• Support for platforms other than Web, Android, and iOS (for example, consoles or native desktop applications). • Online/remote multiplayer mode between players in different locations, unless defined later by the team. • Translation of the game into languages other than Spanish. • Monetization or commercial distribution of the product.

• Brand identity (name, logo, slogan) and optional mascot (female Jack Russell dog). • Mode for 1 to 4 players per team. • Age restriction 14+ and content free of violence, pornography, and cybercrime. • Documentation: instructions, user manual, installation manual, maintenance manual, mockups, and prototypes.	• Additional levels beyond the initial 6, in this first version. • Integration with social media or other external systems not included in the initial requirements.
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8. Assumptions

List of the most important assumptions to highlight for the project, each presented in its own table:

Assumption 1	It is assumed that instructor Yuely Adriana Arce will act as the validating client for the project deliverables.
Assumption 2	It is assumed that SENA will provide the equipment, laboratories, and software licenses necessary for development.
Assumption 3	It is assumed that the 4 members of the Gameplay team will be available throughout the entire development of the project.
Assumption 4	It is assumed that the game's development engine will be Unity, in line with the teams already formed for that area.
Assumption 5	It is assumed that the mascot (female Jack Russell dog) is an optional, not mandatory, element for the first version of the game.
Assumption 6	It is assumed that the minimum of 1000 users refers to target/potential users of the product, not to a concurrent server load requirement.
Assumption 7	It is assumed that accessibility testing will be validated with test users and/or evaluation tools, without requiring official external certification.

9. Constraints

Type of Constraint	Description
End date	The project must be completed and delivered no later than November 19, 2026.
Milestone dates	Intermediate milestones must be met within the academic term schedule, with no possibility of extending the final deadline.
Resource availability	Development depends on the laboratories, equipment, and training schedules available at SENA.
Skill availability	The team must work with the tools and technologies already defined in the training program (Unity, office productivity tools, Git/GitHub version control).
Budget	The project is educational in nature and has no budget or external funding allocated.

10. General Risk Factors

The 15 most important risk factors that could affect the project, along with their impact consequence and the owner of the response plan:

#	Risk Factor	Consequence of Impact	Response Plan Owner
1	Delays in the delivery of mockups by the UX/UI team.	Delay in the development of the interface and game screens.	Joseph Gómez (UX/UI)
2	Technical difficulties in integrating the Unity engine.	Delay in building and testing the levels.	Luiyer Gamalie / Michael Ocampo (Unity)
3	Lack of defined internal roles within the Gameplay team.	Duplicated efforts or gaps in work within the team.	Michael Isaza (Gameplay)
4	Scope changes requested by the instructor/client.	Rework and deviation from the planned schedule.	Juan Esteban Montoya (Project Manager)
5	Low availability of team members' time due to other courses or commitments.	Failure to meet established milestones.	Juan Esteban Montoya (Project Manager)
6	Difficulty implementing full accessibility (people with no hands, who are deaf, or who are blind).	Failure to meet a key project requirement.	Joseph Gómez (UX/UI) and Michael Isaza (Gameplay)
7	Failures in the autosave system.	Loss of player progress and poor user experience.	Thomas (Backend)
8	Performance issues on low-end mobile devices.	Poor gaming experience on Android/iOS.	Unity Teams (Luiyer Gamalie / Michael Ocampo)
9	Insufficient testing before final delivery.	Bugs present in the delivered version.	Entire project team
10	Generated content that violates the restrictions (violence, inappropriate language, etc.).	Rejection or low grading of the project for failing to comply with established policies.	Leandro Calvo (Game Design)
11	Delays in defining the brand identity (name, logo, slogan).	Delay in communication materials and project documentation.	Leandro Calvo (Game Design)
12	Poorly managed cross-team dependencies (for example, Gameplay depends on Game Design's levels).	Bottlenecks that delay several work fronts at the same time.	Juan Esteban Montoya (Project Manager)
13	Loss of information or code due to improper use of version control.	Rework and loss of progress already made.	Thomas (Backend) and each team's leaders
14	Difficulty reaching the minimum of 1000 users within the project timeframe.	One of the defined success criteria for the project is not met.	Juan Esteban Montoya (Project Manager)
15	Inconsistencies between the web version and the mobile version of the game.	Different brand and gameplay experience depending on the platform used.	Joseph Gómez (UX/UI)

11. Stakeholders

Individuals and organizations with an interest in the project, or impacted by its outcomes:

Stakeholder	Role / Relationship to the Project	Type of Interest / Impact
Instructor Yuely Adriana Arce	Client / Project Evaluator	Positive — high interest in meeting the requirements
SENA (institution)	Sponsor of the educational project	Positive — interest in the educational outcome of the project
Juan Esteban Montoya	Project Manager / Scrum Master	Positive — responsible for the overall success of the project
Game Design Team	Responsible for levels and narrative	Positive — interested in having their design correctly implemented
UX/UI Team	Responsible for design and interface	Positive — interested in the usability and accessibility of the final product
Unity Teams	Development of the game in the Unity engine	Positive — interested in the technical feasibility of development
Gameplay Team	Design and implementation of game mechanics	Positive — direct interest in the quality of the playable experience
Backend Team	Infrastructure, data, and server logic	Positive — interested in the stability of the scoring and save system
End users (players 14+)	Consumers of the product	Positive — benefit from a playful learning experience
Users with disabilities (no hands, deaf, blind)	Direct beneficiaries of the game's accessibility	Positive — direct impact if accessibility is achieved; negative if it is not
Parents or guardians of underage users (14-17 years old)	Interested in the suitability of the content	Can be positive or negative depending on compliance with content restrictions
Classmates from the other 5 groups in the course	Peer evaluation and feedback	Neutral/positive — academic interest in the outcome

12. Key Milestones

The project's most important milestones, defined according to the work assigned to the Gameplay team:

Date	Milestone
06/08/2026	Project kickoff and approval of the Project Charter.
20/08/2026	Brand identity defined (name, logo, slogan) and base game mechanics outlined.
10/09/2026	Gameplay mechanics design document (difficulty, scoring, levels) finalized.
01/10/2026	Functional prototype of mechanics for levels 1 through 3.
22/10/2026	Gameplay mechanics integrated with the 6 levels developed in Unity.
05/11/2026	Playtesting and adjustment of difficulty levels (easy, intermediate, expert).
12/11/2026	Final Gameplay documentation (contribution to user manual and in-game instructions).
19/11/2026	Final project delivery.

